ALEXANDRIA VINCENT

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EDUCATION

Villanova University, Villanova, PA
Bachelor of Science in Computer Science, Minor in Cybersecurity — Expected May 2027
Master of Science in Computer Science, Five-Year BS/MS Program — Expected May 2028
GPA: 3.88/4.00 | Dean's List, Every Semester
Coursework: Data Structures & Algorithms, Analysis of Algorithms, Cybersecurity, Linear Algebra, Computer Systems

RESEARCH EXPERIENCE

Undergraduate Cybersecurity Research Assistant | Villanova University — *June 2026 – Present*

- Research LLM-assisted intrusion detection under faculty mentorship, investigating explainability and multimodal decision fusion for security telemetry.
- Built an end-to-end Python pipeline aligning TON_IoT Windows host and network-flow telemetry into 21,000+ paired events across seven attack classes; train, calibrate, and evaluate per-modality decision-tree detectors with episode-grouped splits to prevent near-duplicate data leakage.
- Diagnosed overconfident decision-tree predictions that tied on nearly a third of all fusion decisions; designed a probability-calibration method eliminating every tie at no cost in accuracy.
- Designed controlled experiments evaluating GPT-4o decision fusion against mechanical and learned baselines over 19,000+ API calls; measured sensitivity to prompt wording and input ordering, and reported results with bootstrap confidence intervals and McNemar significance testing.
- Co-authoring a conference paper (in preparation) on when LLM-assisted fusion helps or hurts intrusion detection; own methodology, results, and reproducibility documentation.

SECURITY ENGINEERING EXPERIENCE

IT Security Engineering Intern | EPC (Entry Point Communications) — *May 2026 – Sept. 2026*

- Designed, built, and tested a modular Python penetration-testing framework for authorized kiosk security assessments, automating host discovery, TCP port scanning, and service enumeration with configuration-driven policies, logging, and structured reporting.
- Executed authorized network and wireless testing against physical kiosk hardware using Linux security tools; presented findings to engineers and documented configuration, execution, and handoff procedures.
- Evaluated security-awareness training platforms to support SOC 2 compliance initiatives.

Software Engineering Intern | EPC (Entry Point Communications) — *Summer 2025; Dec. 2025 – Jan. 2026*

- Developed a Python barcode-scanning system integrated into EPC production software, using a plugin-based architecture with structured output and logging; debugged and tested with Git and CI/CD workflows.

TECHNICAL SKILLS

Security & AI: Penetration Testing, Network Scanning (Nmap), Intrusion Detection, LLM-Assisted Detection, Model Evaluation, Probability Calibration, Statistical Hypothesis Testing, Decision Trees, Digital Forensics
Languages: Python, Java, C#, C++, JavaScript, SQL, HTML, CSS
Tools & Platforms: scikit-learn, pandas, Git, GitHub, Linux, CI/CD Pipelines, Unity

PROJECTS

XR Sports Prediction and Trivia Platform | Villanova University — *Jan. 2026 – Present*

- Develop a Unity/C# extended-reality app for live sporting events with predictions, trivia, and leaderboards on Magic Leap hardware.

LEADERSHIP

NCAA Division I Student-Athlete, Women's Soccer | Villanova University — *Aug. 2023 – Present*

- Train and compete ~20 hours per week while maintaining a 3.88 GPA and Dean's List every semester.

Representative, Student-Athlete Advisory Committee (SAAC) | Villanova University — *Oct. 2023 – Present*

- Represent the Women's Soccer Team to athletic administrators on student-athlete concerns and initiatives.